FInk: An On-Device, Evidence-Gated Retrieval System for Financial-Risk Review of Creator Contracts

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Abstract

Creators sign revenue-sharing contracts under time pressure and with little financial training, and the clauses that move the most money—settlement basis, deductions, recoupment, payment timing—are the easiest to miss. Making review available is not the same as getting it used: a large field study found that the people who most need unbiased financial advice are the least likely to take it. We present FInk, an on-device assistant that reads a creator's contract, detects financial risk across nine cash-flow categories, grounds each flag in a curated corpus of official Korean standard contracts and counseling casebooks through BM25 retrieval, and returns a prioritized review: a 0–100 review-attention score, a recommended review effort, and ranked, source-cited findings, with a local chatbot for follow-up questions. An evidence gate lets only officially grounded clauses raise the score, so unverified signals inform questions but never inflate risk. On synthetic, sanitized fixtures a rule + model hybrid gives the best risk detection (macro-F1 0.92) while keeping false positives low, retrieval returns the correct authority tier and is consistent across Korean and English queries, and a decision-aware ordering covers more financial exposure within a small reading budget than a severity-only baseline; the whole pipeline runs locally with no network calls. FInk is decision support for what to check before signing, not legal advice or a verdict. FInk is available at https://fink.seonukkim.com.

1. Introduction

The creator economy runs on contracts. Webtoon artists, illustrators, and writers sign agreements that fix how revenue is split, what is deducted before they are paid, when the money actually arrives, and who keeps the secondary rights. Those terms decide a creator's real cash flow, often more than the headline rate does. The people signing them usually have no finance or legal background and little time before a deadline.

The clauses with the largest financial consequence are also the easiest to skip. A recoupment clause, a vague deduction basis, or a ninety-day settlement delay can quietly move a large share of income, yet each sits among many routine terms and reads as boilerplate. Signing is close to irreversible: afterward, renegotiation is hard. Professional review exists, but offering advice is not the same as people using it. In a field study of roughly eight thousand retail investors, the clients who most needed unbiased advice were the least likely to obtain it, and the few who did barely followed it; the authors conclude that unbiased advice is necessary but not sufficient (Bhattacharya et al., 2012). A tool that lowers the cost of knowing what to ask before signing targets that uptake gap, not only the availability gap. Privacy makes the problem sharper: a draft contract is sensitive personal and business information, and uploading it to a cloud service to be read is itself a disclosure many creators would rather avoid.

We present FInk, an on-device assistant for financial-risk review of creator contracts. FInk reads a pasted or photographed contract, splits it into clauses, detects financial risk across nine cash-flow categories—from settlement transparency to production-cost burden—and grounds each flag by retrieving supporting passages from a curated corpus of official Korean standard contracts and counseling casebooks. It then reports a prioritized review: a 0–100 review-attention score, a three-level recommended review effort, and a ranked list of findings, each naming its clause and the sources behind it. A small on-device chat model answers follow-up questions over the same grounded material, and the review can be saved as a one-page

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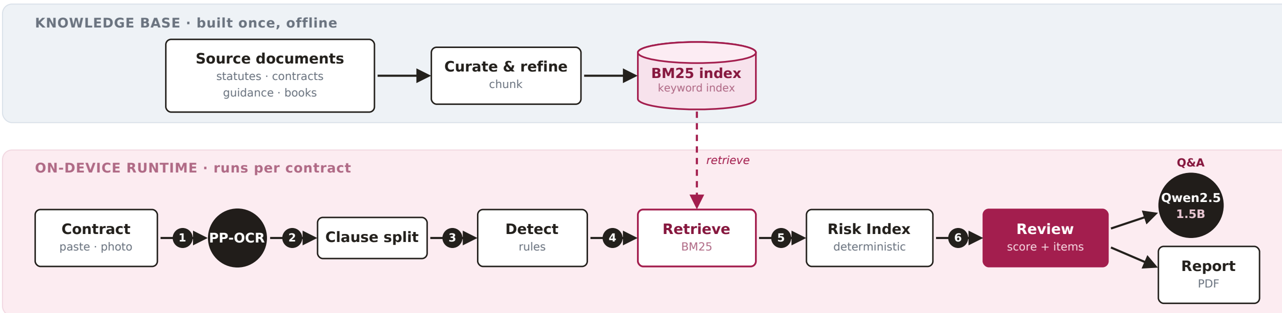

Figure 1. FInk pipeline. An offline knowledge base of curated Korean sources is built once and indexed for retrieval. At runtime, per contract and on device, the text is split into clauses, financial risk is detected, each signal is grounded by BM25 retrieval against the knowledge base, and a deterministic review-attention score and ranked findings are produced; a small local model answers follow-up questions, and the review exports as a brief.

brief.

The choice that holds FInk together is an evidence gate. Only clauses supported by official, score-eligible sources contribute to the score; signals taken from practice books or unverified text can still raise a question, but they never raise the risk number. The score is a deterministic function rather than a learned black box: a saturating per-category severity combined into a bounded index, with weights drawn from standard risk-assessment practice. This keeps FInk conservative by construction and keeps its output readable as a review priority, not a ruling.

Real creator contracts are private, so we evaluate on synthetic, sanitized fixtures and report every number as measured-on-fixture rather than a generalized claim. On these fixtures a rule + model hybrid detects financial risk best while keeping false positives low; retrieval returns the correct authority tier and agrees across Korean and English queries; a decision-aware ordering covers more financial exposure within a small reading budget than a severity-only baseline; and the pipeline runs locally with no outbound calls. Our contributions are:

- a financial-risk taxonomy for creator contracts and an evidence-gated scoring design that separates what to review from what is verified, so unverified signals never inflate the risk score;
- an on-device retrieval-augmented pipeline that grounds creator-contract risk in official Korean financial and legal sources and answers follow-up questions without the contract leaving the device; and
- a decision-focused evaluation, on synthetic fixtures, that measures whether FInk's ordering helps a creator cover financial exposure under a limited reading budget, not only whether clauses are classified correctly.

2. Related Work

Retrieval-augmented generation for financial documents. Retrieval-augmented generation answers from retrieved evidence rather than from parameters alone (Lewis et al., 2020), with BM25 (Robertson & Zaragoza, 2009) and dense retrieval (Karpukhin et al., 2020) as the standard retrievers. Financial question answering has moved toward realistic retrieval. FinDER shows that real financial queries are short and jargon-heavy, so the hard part is retrieval rather than reading a context that has already been handed to the model (Choi et al., 2025a); FinAgentBench frames retrieval as multi-step, first choosing the document type and then the passage (Choi et al., 2025b). FInk applies evidence-gated retrieval to a different financial document, the creator's own contract, over a small curated Korean corpus, and runs the whole loop on device, where a heavy dense retriever is impractical and a sparse BM25 index is a deliberate fit rather than a fallback.

Decision-focused and responsible financial AI. Decision-focused learning argues that a predictor should be judged by the decision it supports, not only by predictive error (Elmachtoub & Grigas, 2022). Evaluation of financial language models is fragile: Kong et al. (2026) catalog biases—among them narrative and objective bias—that let fluent, confident output look like a result, and warn against treating persuasion as evidence. And availability is not uptake: unbiased advice was necessary but not sufficient in the field study above (Bhattacharya et al., 2012). FInk is decision-focused in spirit—it orders findings by financial exposure and is evaluated on whether that ordering covers more exposure under a reading budget—but it does not train end to end through an optimizer. The evidence gate and the explicit not-a-verdict framing are our response to the narrative and objective biases that make confident financial outputs hard to trust.

On-device and privacy-preserving document AI. Small instruction-tuned models such as Qwen2.5 (Qwen Team, 2024) and lightweight OCR such as PP-OCR (Du et al., 2020) make local document understanding workable on commodity hardware. Earlier on-device work usually targets latency or cost. FInk instead treats locality as the privacy property itself—a draft contract never leaves the device—and accepts the resulting capability ceiling, a 1.5B chat model, as a design constraint rather than a limitation to hide.

3. Design Goals

The problem and the responsible-AI concerns above give five goals; every later design decision points back to one of them.

- G1. Decision support, not a verdict. The output is a review priority and a set of questions to ask, never a ruling on legality, fraud, validity, fairness, or guaranteed loss.
- G2. Evidence-gated scoring. Only official, score-eligible sources can raise the score. Unverified signals inform questions only.
- G3. On device by default. Ingestion, OCR, retrieval, scoring, and the chat model run locally; no contract text leaves the device.
- G4. Prioritized and actionable. The creator receives an ordering—what to check first and why—that fits a small reading budget, plus a portable brief.
- G5. Korean-canonical and bilingual. Korean source text is canonical; English is a retrieval and reading aid, marked as generated and not an equivalent legal reading.

4. System

FInk has two phases (Figure 1): an offline knowledge base built once, and an on-device runtime that runs per contract.

4.1. Knowledge base and authority tiers

The knowledge base is curated from real Korean sources. Official material—sector standard contracts, official contracting guides, and public counseling casebooks—is score-eligible and tagged by authority tier (A0–A2). Two creator-law reference books are kept as practice context only: they are distilled into checkpoint summaries and never contribute to the score (tiers B/C). This split is the mechanism behind G2. Official sources were preprocessed into Markdown with Claude Opus 4.8 and the two books with ChatGPT 5.5 Pro, then indexed with BM25 (Robertson & Zaragoza, 2009); the raw source text is not redistributed, and only distilled, source-typed summaries are committed. We use BM25 rather than a dense retriever for three reasons that fit the setting: the corpus is small and curated, exact Korean legal terms matter, and a sparse index runs on device with no model to host.

4.2. From contract to clauses

The creator pastes text or uploads a photo or PDF. Images and scans pass through PP-OCR in its Korean configuration (Du et al., 2020), a small CPU OCR model; this is one input path, not the focus of the system, and extraction quality is reported only in the appendix. The text is then segmented into clauses, which are the unit of detection and grounding.

4.3. Financial-risk detection

Each clause is checked against nine cash-flow risk categories, F1–F9: settlement transparency and audit rights (F1); revenue base and deductions (F2); payment and cash-flow timing (F3); minimum guarantee and recoupment (F4); intellectual property and secondary-rights monetization (F5); term, exclusivity, and opportunity cost (F6); termination, liability, and penalties (F7); scope creep and production-cost burden (F8); and a residual category (F9). These cover where a creator-contract clause most often moves cash. Detection runs in three arms: a set of deterministic rules, a small local classifier, and their union (the hybrid). We ship the rule arm by default because its decisions are inspectable, and report all three.

4.4. Evidence-gated grounding

For each detected signal, BM25 retrieves supporting passages from the knowledge base. A signal becomes score-eligible only when an official (A0–A2) source backs it; otherwise it is kept as a practice-informed note that can raise a question but not the score. The not-a-verdict stance (G1) is therefore enforced in the data path, not only in the wording: the score cannot move on text that no official source supports.

4.5. Review-attention score

The score is deterministic. For each category c we map the grounded severity x_c to a per-category contribution $g_c(x_c)$ that saturates—extra severity past a point adds little—following standard risk-assessment practice (International Organization for Standardization, 2018; International Electrotechnical Commission, 2018), and is capped at a category weight w_c. The contributions are summed into a single review-attention index $R = \sum_c g_c(x_c)$ that we keep in $[0, 100]$ (Proposition 1). The weights encode two ideas from decision theory: the same nominal exposure hurts more when it is less recoverable or more uncertain, which is risk aversion (Pratt, 1964), and a single attention number must trade off several categories that are not directly comparable, which is multi-attribute value (Keeney & Raiffa, 1976). We keep how strongly a signal is supported separate from how severe it is, in the spirit of GRADE (Guyatt et al., 2008): evidence strength gates eligibility, severity sets magnitude. The index maps to a three-level recommended review effort—light check, careful check, or professional review recommended.

4.6. On-device chat and brief

A small instruction model, Qwen2.5-1.5B-Instruct in a 4-bit build (Qwen Team, 2024), answers follow-up questions over the grounded findings. It is prompted to stay on the retrieved material and to keep the review framing rather than issue a ruling. The review exports as a one-page brief through the browser's own print path, with no remote document service, consistent with G3 and G4.

5. Evaluation

Real contracts are private, so all evaluation uses synthetic, sanitized fixtures with a frozen split created once and never edited; tuning used a separate development split. Every number is measured-on-fixture and is not a generalized performance claim. We focus the evaluation on the financial-AI parts of the system; OCR and extraction numbers are in Appendix C.

5.1. Financial-risk detection

Table 1 compares the three detection arms over F1–F9. The hybrid gives the best macro-F1 while holding the benign false-positive rate at the rule level; the model arm alone reaches similar recall but doubles false positives on benign clauses. On this fixture, combining inspectable rules with the model detects risk best without becoming trigger-happy on harmless terms.

Table 1. Financial-risk detection over categories F1–F9 (synthetic fixture). Higher macro-F1 is better; lower benign false-positive rate (FPR) is better.

Arm	Macro-F1	Benign FPR
Rule only	0.83	0.25
Model only	0.86	0.50
Hybrid	0.92	0.25

5.2. Evidence grounding

On the retrieval fixture, authority-tier correctness is 1.00: practice (B/C) sources are never returned as score-eligible, which is the property G2 depends on. Recall@3 and Recall@5 are both 1.00, Korean and English versions of a query resolve to the same evidence (1.00 consistency), and cited spans overlap the gold spans at 0.85. The gate behaves as designed on the fixture and is bilingual-consistent.

5.3. Decision quality

The decision-focused question is whether FInk's ordering helps a creator cover financial exposure under a small reading budget, not only whether clauses are labeled correctly (Elmachtoub & Grigas, 2022). We compare an exposure-aware ordering against a severity-only baseline on a factorial fixture with hidden oracle exposure weights, and measure oracle exposure coverage at k (OEC@k): the share of total financial exposure captured by the top-k clauses in a given order (Table 2). Reading in FInk's exposure-aware order covers more exposure within one to three clauses than reading by severity. When the evidence gate is on, coverage is lower because only officially grounded clauses are allowed to count; this is the conservative trade-off the gate is meant to make—it trades coverage for trustworthiness. In a separate cost-sensitive fixture with a fixed trigger threshold, the verification trigger recovers 0.67 of the consequential clauses at a false-trigger rate of 0.33, against a synthetic trade-off between the cost of a missed exposure and the effort of verification; this frames review as a cost-sensitive decision rather than plain classification, in the spirit of the course's risk-scoring material.

5.4. Scoring correctness and runtime

The scoring module passes all registered formula checks (13/13 unit cases and 10/10 financial-scenario

Table 2. Decision quality: oracle exposure coverage at k (OEC@k), higher is better, on the synthetic factorial fixture. Exposure-aware ordering covers more exposure under a small reading budget than severity-only; the evidence gate lowers coverage in exchange for only counting verified clauses.

Evidence gate	Ordering	OEC@1	OEC@3
Off	Severity baseline	0.69	0.93
Off	Exposure-aware	0.81	1.00
On	Severity baseline	0.41	0.55
On	Exposure-aware	0.53	0.62

cases), and the review-attention index stays inside $[0, 100]$ by construction (Proposition 1). On the runtime fixture, FInk made zero network attempts across accepted runs, leaked no contract text or file paths across six log and export surfaces, and ran end to end in well under a second with a peak memory of tens of megabytes. The on-device privacy property of G3 holds on the fixture.

6. Discussion

Decision support, not automation. FInk reports a priority rather than a verdict for a reason tied to the task. Signing is near-irreversible and the money at stake is the creator's own income, so the failure that matters most is a confident wrong reading the creator acts on. Keeping the output a review ordering, and routing the strongest cases to "professional review recommended," keeps the person in the decision and uses AI to lower the cost of knowing what to ask. This is the uptake gap from Bhattacharya et al. (2012) addressed at the level of the tool rather than the advisor.

Evidence-gating as a conservative stance. The narrative and objective biases that Kong et al. (2026) warn about are exactly what an eager contract-reading model would produce: fluent, confident readings that look like legal conclusions. The evidence gate is the structural answer. A signal can raise the score only if an official source backs it, so the score cannot be inflated by persuasive but ungrounded text. The price is visible in our own numbers—coverage drops when the gate is on (Table 2)—and we accept it as the cost of not overclaiming.

Privacy against capability. Running on device buys privacy and costs capability. A 1.5B chat model grounded by retrieval is weaker than a large hosted model, and in practice its free-form answer quality was the main limit we hit. Retrieval narrows the gap by holding answers to retrieved evidence, but does not close it. We treat this as the honest price of G3 rather than something to paper over, and it sets the agenda for future work: better grounding and calibration for small local models.

What changes for the creator. FInk shifts the pre-signing step from "read everything or read nothing" to "read the few clauses that move the most money, with the questions to ask attached." The exported brief carries that into a negotiation.

7. Limitations

The evaluation is on synthetic, sanitized fixtures; there is no study with real creators or real contracts, and no number here should be read as generalizing beyond the fixture. The on-device chat model has a real answer-quality ceiling that we did not fully calibrate, which we accept as a consequence of G3. The knowledge base is Korean and centered on webtoon and copyright material; one intended source, an arts-welfare casebook, was unavailable and is not in the live knowledge base, so some real risks may go ungrounded. Deterministic detection misses paraphrases the rules do not anticipate. FInk is a review aid, not a ruling on legality, fraud, validity, fairness, or loss, and important decisions still need a professional. As a single-author course project, the depth of the evaluation matches that scope.

8. Conclusion

FInk is an on-device, evidence-gated retrieval-augmented assistant that turns a creator's contract into a prioritized financial-risk review and answers follow-up questions, without the contract leaving the device. On synthetic fixtures a rule + model hybrid detects financial risk best while staying conservative on benign clauses, the evidence gate returns the correct authority tier, and a decision-aware ordering covers more exposure under a reading budget than a severity baseline. The design keeps the creator in the decision and treats privacy and conservatism as first-class constraints. The next step is to move from synthetic fixtures to a study with real creators and to strengthen grounding for the on-device model.

Software, Data, and Use of AI Tools

The implementation uses PaddleOCR PP-OCR for optional image and PDF input, a BM25 index for retrieval, a deterministic module for the review-attention score, and Qwen2.5-1.5B-Instruct (a 4-bit GGUF model run through llama.cpp) for on-device chat, served by a FastAPI application. The knowledge base was built from official Korean sources (sector standard contracts, contracting guides, and public counseling casebooks) and two creator-law reference books; official sources were preprocessed into Markdown with Claude Opus 4.8 and the two books with ChatGPT 5.5 Pro, then indexed, and the raw source text is not redistributed. The system was built with AI coding assistants (OpenAI Codex and Anthropic Claude Code) under a specification-driven loop, and this paper was drafted with LLM assistance. All AI-assisted outputs were checked by the author through automated gates, tests, and manual review, and every reference was verified by hand. The author is responsible for all submitted material.

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A. Review-Attention Score Bound

Let the review-attention index be $R = \sum_{c=1}^{9} g_c(x_c)$, where $x_c \geq 0$ is the grounded severity in category c and g_c is the per-category contribution. Each g_c is non-decreasing, saturating, and capped at a category weight $w_c \geq 0$, so that $0 \leq g_c(x_c) \leq w_c$ for all x_c. The weights are chosen with $\sum_c w_c = 100$.

Proposition 1. For any non-negative severities $(x_1, \ldots, x_9)$, the index satisfies $0 \leq R \leq 100$.

Proof. Each term is bounded, $0 \leq g_c(x_c) \leq w_c$. Summing over c gives $0 \leq \sum_c g_c(x_c) \leq \sum_c w_c = 100$, that is $0 \leq R \leq 100$. Two degenerate cases are guarded in code: a non-positive saturation rate is rejected so that g_c stays well defined, and a non-positive weight sum is rejected so the normalization to $\sum_c w_c = 100$ is valid. □

The unit and financial-scenario suites in Section 5 exercise this form on registered worked examples; they check that contributions saturate, that the evidence gate zeroes the contribution of an ungrounded signal, and that R stays in range under the registered cases.

B. Knowledge-Base Sources

Score-eligible official sources include sector standard contracts and annotations for cartoons and webtoons (Korea Creative Content Agency, 2025), standard contracts for the assignment and licensing of authors' economic rights (Ministry of Culture, Sports and Tourism (Korea) & Korea Copyright Commission, 2018), a fair-contracting guide for the sector (Korea Creative Content Agency, 2023), a copyright counseling casebook (Korea Copyright Commission, 2025), and a content-dispute mediation casebook (Content Dispute Resolution Committee (Korea), 2024). Non-scoring practice context comes from two creator-law reference books (boo, b;a), which are distilled into checkpoint summaries and never raise the score. One intended official source (an arts-welfare rights-infringement casebook) was unavailable in the corpus and is not part of the live knowledge base.

C. OCR and Extraction on Fixtures

For completeness, the optional OCR and value-extraction path was measured on the synthetic camera/OCR fixture: character error rate 0.053, word error rate 0.125, exact-match accuracy 1.00 on extracted money, percentage, date, and duration values, and clause-segmentation quality 1.00. These describe input quality on the fixture and are not the contribution of this paper.

D. Reproduction

The local application runs with uv sync --extra web followed by uv run fink-web, then opening the printed local address. The project page and a runnable demo are at https://fink.seonukkim.com.